

Customer Purchase Prediction

1. Project Overview

This project aims to predict the type of purchase intent of an online customer — specifically whether a purchase is Need-based, Wants-based, Planned, or Impulsive — using their behavioral and engagement data. A supervised machine learning approach is employed, training a Random Forest Classifier on the E-Commerce Consumer Behavior Analysis Dataset.

The core goal is to enable e-commerce platforms to better understand customer decision-making patterns, optimize marketing strategies, and personalize user experiences in real time through a Flask-based web application.

2. Objectives

- To analyze customer behavior data from e-commerce platforms
- To preprocess and clean the dataset for effective modeling
- To build a machine learning model to predict purchase intent
- To evaluate the model using standard performance metrics
- To identify important factors influencing customer decisions
- To develop a web interface for real-time prediction

3. Target: Purchase Intent Classes

The model classifies each customer session into one of four purchase intent categories as described below:

Purchase Intent	Description	Decision Time
Need-based	Customer purchases driven by necessity or essential requirements	High
Wants-based	Purchases motivated by desires, trends, or lifestyle aspirations	Medium
Planned	Pre-decided purchases with prior research and budget allocation	High
Impulsive	Spontaneous purchases triggered by promotions, ads, or emotions	Low-Medium

4. Input Features

The following features are extracted from the dataset and used as inputs to the classification model:

- Age, Gender
- Income Level
- Frequency of Purchase
- Product Rating
- Social Media Influence
- Customer Satisfaction
- Engagement with Ads
- Time to Decision

5. Technologies Used

The project is built using a modern Python-based stack for data science and web deployment:

- Python
- Flask
- Scikit-learn
- Pandas
- NumPy
- Matplotlib
- HTML and CSS

6. About the Model

Algorithm

The project uses the **Random Forest Classifier** — an ensemble learning method that constructs multiple decision trees during training and outputs the class that is the mode of the individual trees' predictions. It is well-suited for tabular classification tasks with mixed feature types.

Key Characteristics

- Handles both categorical and numerical features effectively
- Robust to overfitting due to ensemble averaging
- Provides feature importance scores for interpretability
- Supports probability-based predictions for each intent class
- Performs well on imbalanced datasets with proper tuning

7. Evaluation Metrics

Model performance is assessed using the following metrics to ensure reliability and generalizability:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score
- Confusion Matrix

8. Web Interface

The trained model is deployed via a Flask web application, providing a user-friendly interface for real-time purchase intent prediction.

Features of the Web Interface

- Input form for all nine behavioral and engagement features
- Real-time prediction of purchase intent category
- Probability scores displayed for each of the four intent classes
- Visualization of feature importance chart
- Class distribution graph for exploratory analysis
- Clean, responsive design using HTML and CSS

